```
# 1) To write a Python program to add two numbers entered by the user.
num1 = float(input("Enter first no.: "))
num2 = float(input("Enter second no.: "))
sum = num1 + num2
print("Sum :", sum)
```

```
Enter first no.: 3
Enter second no.: 5
Sum : 8.0
```

```
# 2) Program to interchange the values of two variables
a = input("Enter first value: ")
b = input("Enter second value: ")
a, b = b, a
print("After swapping:")
print("First value:", a)
print("Second value:", b)
```

```
Enter first value: 3
Enter second value: 6
After swapping:
First value: 6
Second value: 3
```

```
# 3) To write a Python program to find the largest among three numbers using conditional statements.
a = int(input("Enter first no.: "))
b = int(input("Enter second no.: "))
c = int(input("Enter third no.: "))
if a >= b and a >= c:
    print("Largest =", a)
elif b >= a and b >= c:
    print("Largest =", b)
else:
    print("Largest =", c)
```

```
Enter first no.: 5
Enter second no.: 9
Enter third no.: 3
Largest = 9
```

```
#4. To write a Python program to check whether a given number is even or odd.
n = int(input("Enter a number: "))
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

```
Enter a number: 75
Odd
```

```
#5. To write a Python program to determine whether a number is positive, negative, or zero.
n = int(input("Enter a number: "))
if n > 0:
    print("Positive")
elif n < 0:
    print("Negative")
else:
    print("Zero")
```

```
Enter a number: 24
Positive
```

```
#6. To write a Python program to check whether a given year is a leap year.
year = int(input("Enter year: "))
if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```

```
Enter year: 2000
Leap Year
```

```
#7. To write a Python program to perform basic arithmetic operations using operators.
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print("Addition =", a + b)
print("Subtraction =", a - b)
print("Multiplication =", a * b)
print("Division =", a / b)
```

```
Enter first number: 23
Enter second number: 76
Addition = 99
Subtraction = -53
Multiplication = 1748
Division = 0.3026315789473684
```

```
#8. To write a Python program to display the multiplication table of a given number.
n = int(input("Enter a number: "))
for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

```
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

```
#9. To write a Python program to calculate the factorial of a number using a loop.
n = int(input("Enter a number: "))
fact = 1
for i in range(1, n + 1):
    fact = fact * i
print("Factorial =", fact)
```

```
Enter a number: 3
Factorial = 6
```

```
#10. To write a Python program to find the sum of the first N natural numbers.
n = int(input("Enter No.: "))
sum = 0
for i in range(1, n + 1):
    sum = sum + i
print("Sum =", sum)
```

```
Enter No.: 5
Sum = 15
```

```
#11. To write a Python program to print numbers from 1 to N using a for loop.
n = int(input("Enter No.: "))
for i in range(1, n + 1):
    print(i)
```

```
Enter No.: 6
1
2
3
4
5
6
```

```
#12. To write a Python program to print numbers in reverse order from N to 1.
n = int(input("Enter No.: "))
for i in range(n, 0, -1):
    print(i)
```

```
Enter No.: 6
6
5
4
3
2
1
```

```
#13. To write a Python program to print all even numbers up to a given limit.
n = int(input("Enter limit: "))
for i in range(1, n + 1):
    if i % 2 == 0:
        print(i)
```

```
Enter limit: 10
2
4
6
8
10
```

```
#14. To write a Python program to reverse the digits of a given number.
n = int(input("Enter a number: "))
rev = 0
while n > 0:
    rev = rev * 10 + n % 10
    n = n // 10
print("Reversed number =", rev)
```

```
Enter a number: 12
Reversed number = 21
```

```
#15. To write a Python program to count the number of digits in a given integer.
n = int(input("Enter a number: "))
count = 0
while n > 0:
    count += 1
    n = n // 10
print("Digits =", count)
```

```
Enter a number: 24
Digits = 2
```

```
#16. To write a Python program to calculate the sum of digits of a given number.
n = int(input("Enter a number: "))
sum = 0
while n > 0:
    sum += n % 10
    n = n // 10
print("Sum of digits =", sum)
```

```
Enter a number: 23
Sum of digits = 5
```

```
#17. To write a Python program to check whether a given number is prime.
n = int(input("Enter a number: "))
if n > 1:
    for i in range(2, n):
        if n % i == 0:
            print("Not Prime")
            break
    else:
        print("Prime")
else:
    print("Not Prime")
```

```
Enter a number: 34
Not Prime
```

```
#18. To write a Python program to generate the Fibonacci series up to N terms.
n = int(input("Enter number of terms: "))
a = 0
b = 1
for i in range(n):
    print(a)
    c = a + b
    a = b
    b = c
```

```
Enter number of terms: 5
0
1
1
2
3
```

```
#19. To write a Python program to check whether a string is a palindrome.
s = input("Enter a string: ")
if s == s[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

```
Enter a string: 44
Palindrome
```

```
#20. To write a Python program to check whether a given number is an Armstrong number.
n = int(input("Enter a number: "))
temp = n
sum = 0
while temp > 0:
```

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        digit = temp % 10
        sum += digit ** 3
        temp = temp // 10
    if sum == n:
        print("Armstrong Number")
    else:
        print("Not an Armstrong Number")
```

```
Enter a number: 54
Not an Armstrong Number
```